

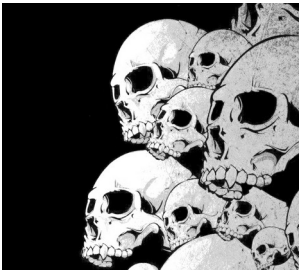
Hydrogen

advanced drum machine for GNU/Linux

<https://www.hydrogen-music.org>

Y. Collette (ycollette.nospam@free.fr)
<https://audinux.github.io>

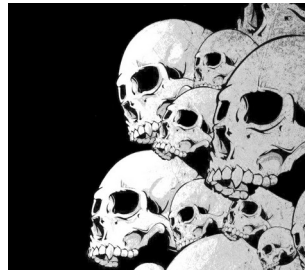




Facility Resources

\$ dnf install hydrogen

3355606kit	Boss_DR-110	EMU_E-DRUM	Korg_ms20+ems_vcs3
AVL-BlackPearl-4A-1.1	Boss_DR-660_taxman_full	EMU_SP1	Korg_minipops35
AVL-BlackPearl-4B-1.1	Boss_DR-660_zapman_full	Ensoniq_Asrx_Acoustic_Full	Korg_minipops7
AVL-BlackPearl-5-1.1	Boss_DR110_full	Ensoniq_Asrx_Electro_Full	Korg_sr120
AVL-RedZep-4-1.0	Boss_DR202_altaira_full	Ernyspercussion	Korg_tx
AVL-RedZep-5-1.1	Boss_DR202_sainthood	Fairlight_cmi-IX_Full	Korg_wavestation
Ace-Tone_Rhythm_Ace	Boss_DR202_zaphod_full	Farfisa	Korg_x5_full
Ace-Tone_Rhythm_Plus	Boss_DR220C	Fortereo	Lightning1024
Akai_MPC-X	Boss_DR550_cymbale	Fricke_mfb5x2	Linn_lm1
Akai_XE8	Boss_DR550_dj	Hamnd_Autovari2	Linn_lm2
Akai_XR10	Casio_MT500	Hamnd_DPM4	Linn_linn-9000
Akai_XR10_Latiny	Casio_MT800	Hardelectro1	Linn_linndrum
Alesis_D4FX_full	Casio_PT30	Hiphop-1	Ltibbits_Rock
Alesis_DM5_circle_full	Casio_PT80	Hiphop-2	Lugenfalscher_beguinesandbeghards
Alesis_DM5_eclectic	Casio_RAP1	Ianhall_SamPlesseet	Lugenfalscher_blitzbox
Alesis_DM5_future_full	Casio_RZ3	Jovica_Tonic_Full	Lugenfalscher_sg1
Alesis_DM5_ghazal_full	Cheetah_MD16_full	K-27_Trash_kit	Lugenfalscher_teatime
Alesis_DM5_space_full	Classic-626	Kawaixd-5kit	Mxr_m185
Alesis_HR16_chaser	Classic-808	Kawai_R1	Matiasreccius_heavymetal
Alesis_HR16_power_full	ColomboAcousticDrumkit	Kawai_R50_Full	Mill-Dums_v.1
Alesis_HR16_skins_full	DeathMetal	Kawai_xd5	Millo_multilayered2
Alesis_HR16_wimpy	EKO_Rhythmbox	Corg_DDD1	Millo_multilayered3
Alesis_SR16	EasternHop-1	Corg_DDD5	Moogish_full
Altemark_gameboy-lsdj	ElectricEmpireKit	Corg_DDM1	Oberheim_dmx
BJAPacific	Electro_Harmonix_Drm15-32	Corg_kpr7	Quasimidi_309
Bdu_Roland_TR606	Emu_Drumulator	Corg_kr5	Rolandjd-990kit
Blackie666_8bit	Emu_Drumulator-alt	Corg_m1	Roland_cr-1000

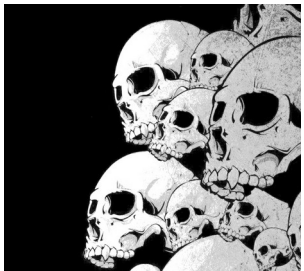


Facility Resources

\$ dnf install hydrogen-drumkit-TD-7kit

Roland_CR79
Roland_D-10_withMOD_full
Roland_D-70
Roland_JD-990_full
Roland_JD800
Roland_MC303
Roland_TB909
Roland_Tr-808_full
SF3007-2011-Set-03
Serge_Modular
Suonho_deconstruction
Synthie-1
TD-7kit
TR808909
Techno-1
Thanvannispen_industrisplode
TicTacShutUp_studio_17-29
VariBreaks
YamahaRX15Kit
YamahaTG-55Kit
YamahaVintageKit
Yamaha_SY22_full
Yamaha_Sy35
Yamaha_TX16W
Yamaha_Tg33_full
Yoursck+Vitriolix_Texturizer





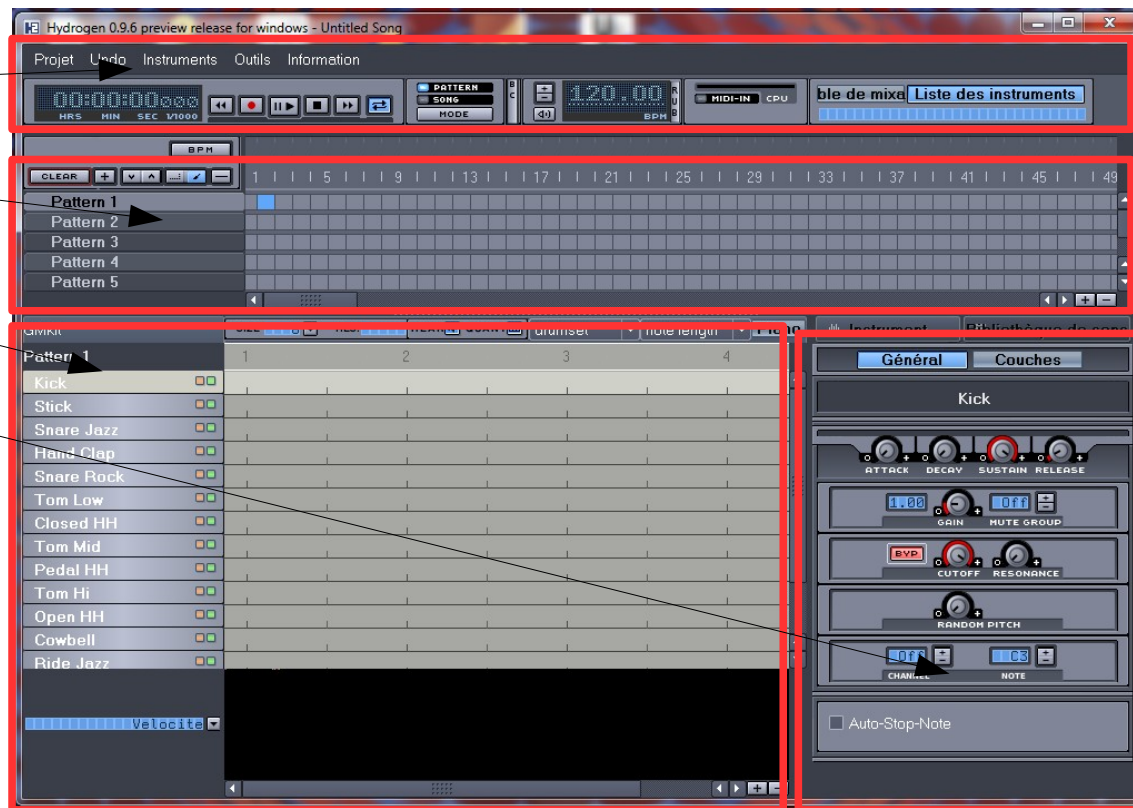
General appearance

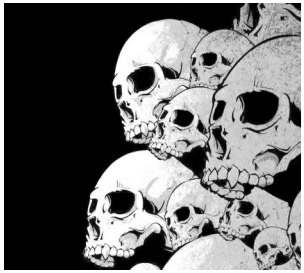
Menus sweep

Patterns sequencer

Patterns

Sound settings





1/2 sound adjustment

ADSR filter associated with each drum sound

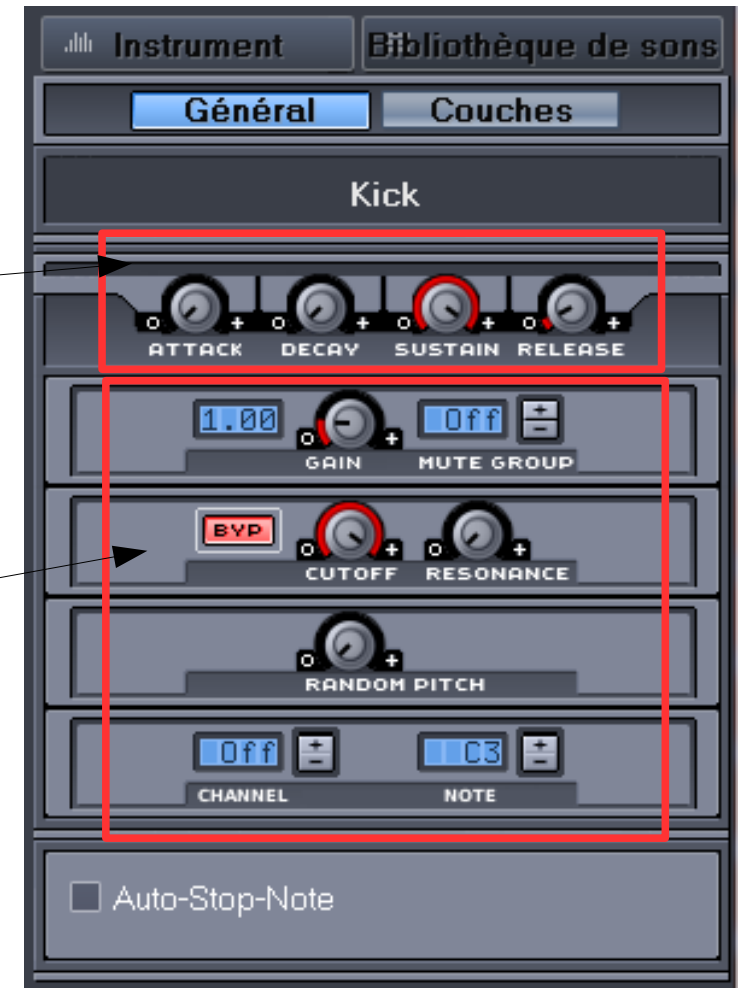
Attack: Attack

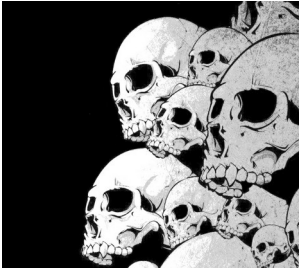
Decay: decrease

Sustain: Maintain

Release: relaxation

Various sound effects

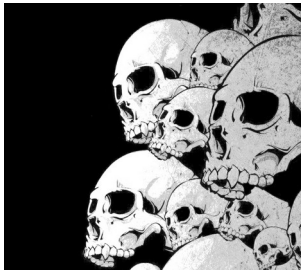




Sound adjustment 2/2

Each drum sound can be a superposition of different sounds.



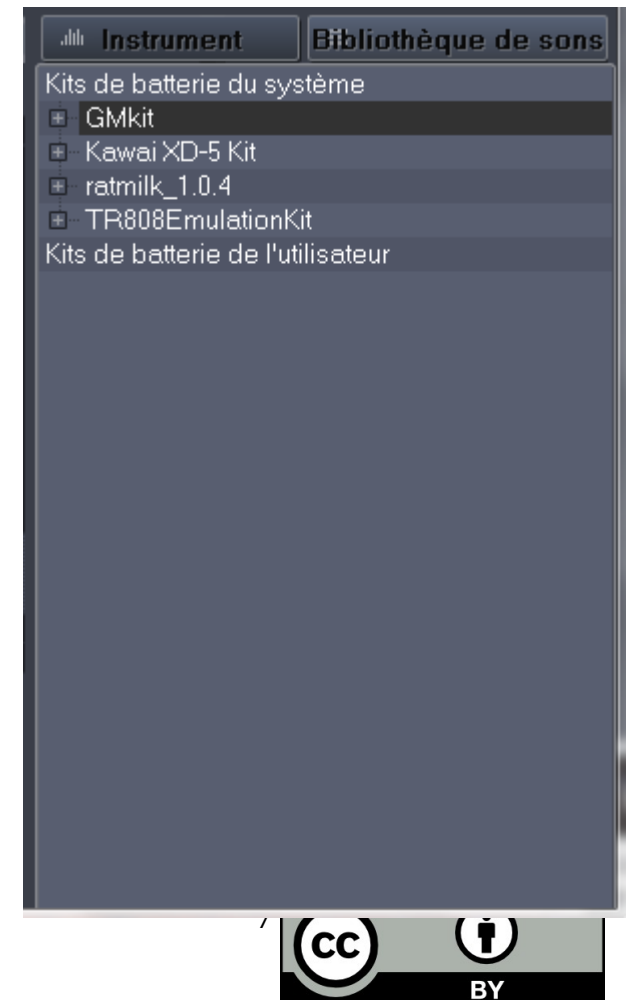


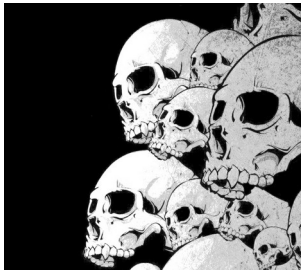
Library selection

Several drum sets exist. Some are downloadable from the site:

<http://sourceforge.net/projects/hydrogen/files/sound%20libraries/>

Version 0.9.7 allows you to download automatically and install new drum sounds.





Import of libraries

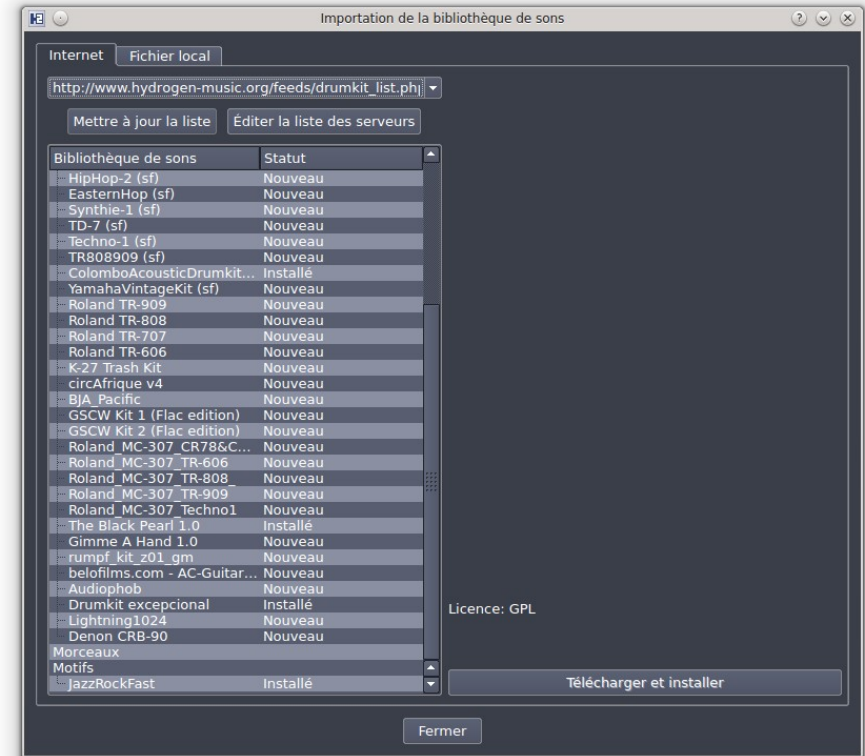
To install a new library, the easiest way is to go through the instrument menu:

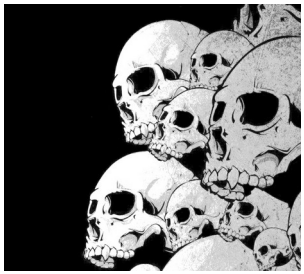
Instruments → Import Library

Click on 'Update List', then select the library you want to install and then click the 'Download and install' button.

Some other libraries are accessible here:

<http://www.hydrogen-music.org/hcms/node/16>





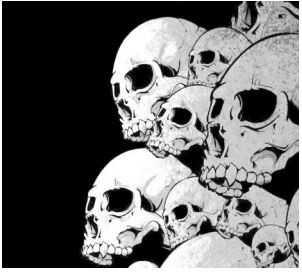
Mixer



Mixing track for each sounds
of the drum

Effect zone of
effects plugins

General
level

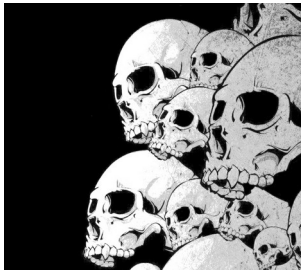


Midi management

Version 0.9.5 manages the Midi Alsa (the Hydrogen input appears in the Alsa Midi section of Qjackctl).

Version 0.9.6 Beta manages the Midi Alsa and Jack (the Hydrogen input may appear to choose from in the Alsa or Jack of Qjackctl section).

```
$ git clone https://github.com/hydrogen-music/hydrogen.git
$ CD Hydrogen
$ mkdir build_tmp
$ cd build_tmp
$ cmake -dcmake_install_prefix =/opt/hydrogen -devel -dcmake_build_type = release ..
$ # Install the missing applications, then relaunch the command above
$ make
$ sudo make install
```



Connect an instrument

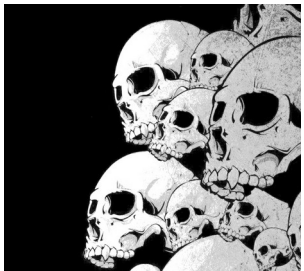
Hydrogen only uses part of the MIDI events to control sounds.
The events managed range from 36 to 59.

To find out that they are generating MIDI events, you have to connect its outing to KMIDIMON. This application makes it possible to display all the MIDI events that go through a given connection.

Then, the hydrogen instruments must be reordered according to the MIDI events of the instrument.

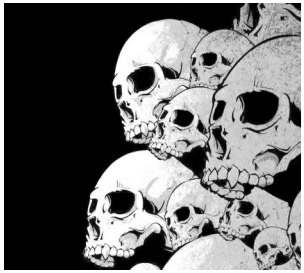
For example, if my instrument generates an event 36 and that we want to associate a snare, then it will be necessary to move the snare in position 1 (so that it is triggered by our MIDI 36 event) in the Drumkit loaded in Hydrogen.

The following table lists the instrument positions with associated MIDI events.



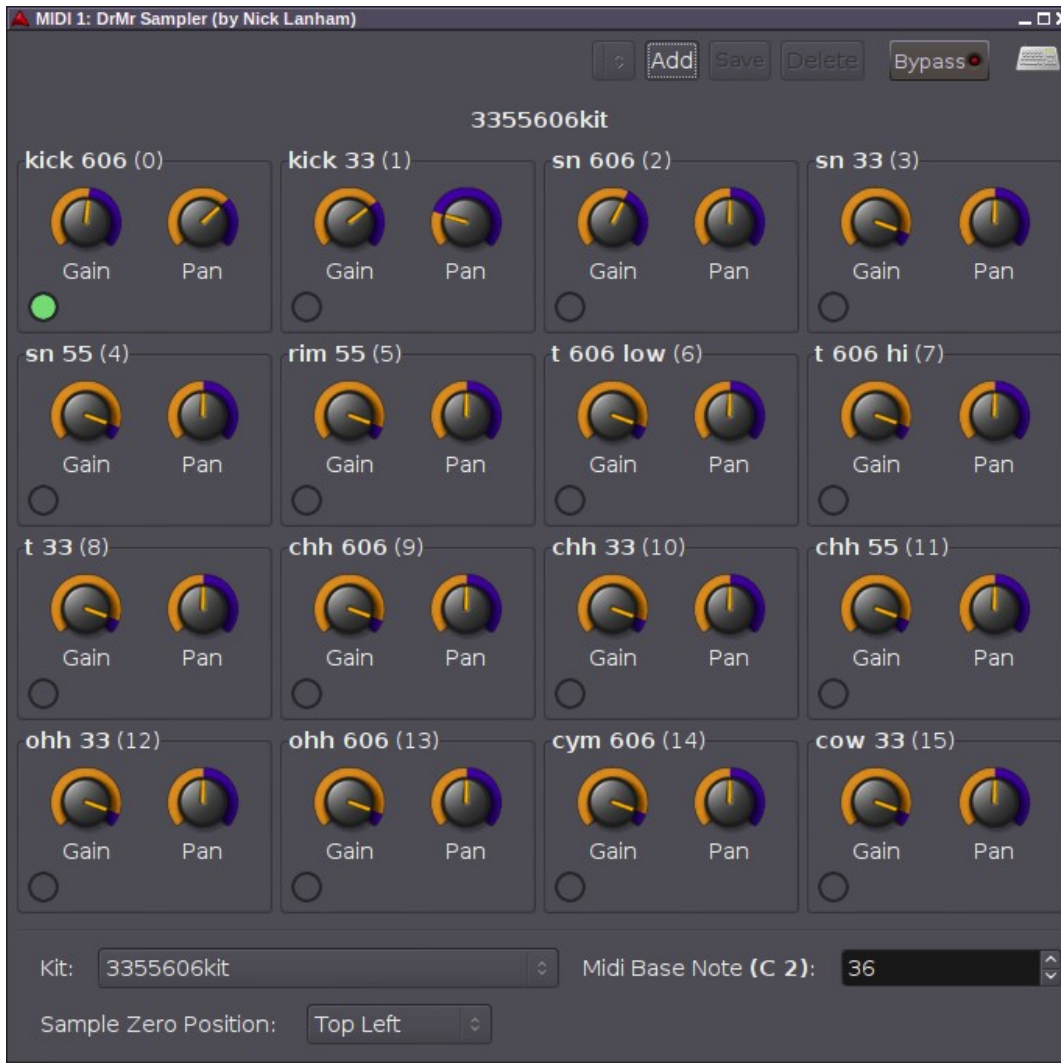
Connect an instrument

Instrument position	MIDI note	PC Keyboard Key	Instrument name (Gmkit)
1	36	Z	Kick
2	37	X	Snare Jazz
3	38	C	Snare Rock
4	39	V	Tom Low
5	40	B	Tom Mid
6	41	N	Tom Hi
7	42	M	Cowbell
8	43	Q	Ride Jazz
9	44	W	Ride Rock
10	45	E	instrument N0. 17 (not assigned yet)
11	46	R	instrument N0. 18 (not assigned yet)
12	47	T	instrument N0. 20 (not assigned yet)
13	48	Y	instrument N0. 22 (not assigned yet)
14	49	U	instrument N0. 24 (not assigned yet)
15	50	S	Stick
16	51	D	Hand Clap
17	52	G	Closed HH
18	53	H	Pedal HH
19	54	J	Popen HH
20	55	2	Crash
21	56	3	Crash Jazz
22	57	5	instrument N0. 19 (not assigned yet)
23	58	6	instrument N0. 21 (not assigned yet)
24	59	7	instrument N0. 23 (not assigned yet)



DrMr

LV2 Plugin LV2 for Hydrogen

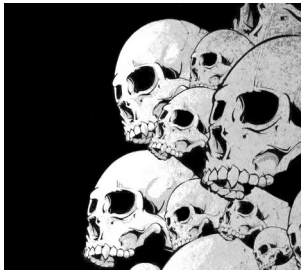


DRMR is a LV2 plugin that allows you to read hydrogen files and generate a sound rendering via MIDI events

<https://github.com/nicklan/drmr>

Other puglins are available:

- Drumrox
- Drumlabooh



An alternative: Drumgizmo

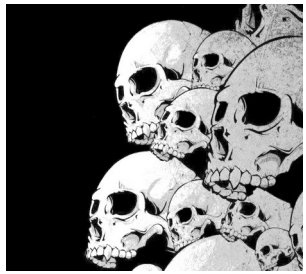


A plugin / executable to simulate a drum.
The current drumkits are rather metal oriented:

- Drskit
- Muldjordkit
- Aasimonster



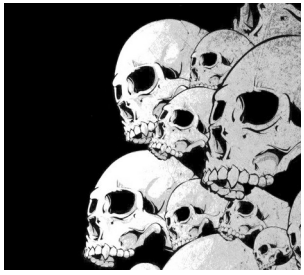
<https://www.drumgizmo.org/wiki/doku.php?id=start>



Drumkit aasimonster



```
$ drumgizmo -i jackmidi -l midimap=midimap.xml -o jackaudio aasimonster.xml
```



Drumgizmo Aasimonster - Micros

This drumkit was recorded using 14 microphones a trigger and the output from a trigger module:

Snare drum was close-mic recorded using two Shure SM57 microphones. One at the top of the snare and one at the bottom. Furthermore the analog output of a DDrum trigger was recorded.

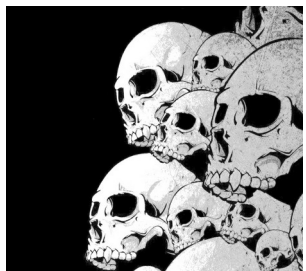
Each of the four toms were close-mic recorded using AKG C518M swaneneck microphones.

The kickdrums were recorded using AKG D112 microphones mounted in the hole. Furthermore the output of a Alesis DM5 drummodule were stored set up with the “Speed Metal” sample. The trigger module were activated using the Axis E-Kit trigger system.

The hihat and ride cymbal were close-mic recorded using cardioid condenser microphones.

The crash cymbals were recorded using RØDE NT5 over head microphones.

The ambiance of the entire kit were recorded using cardioid condenser microphones.

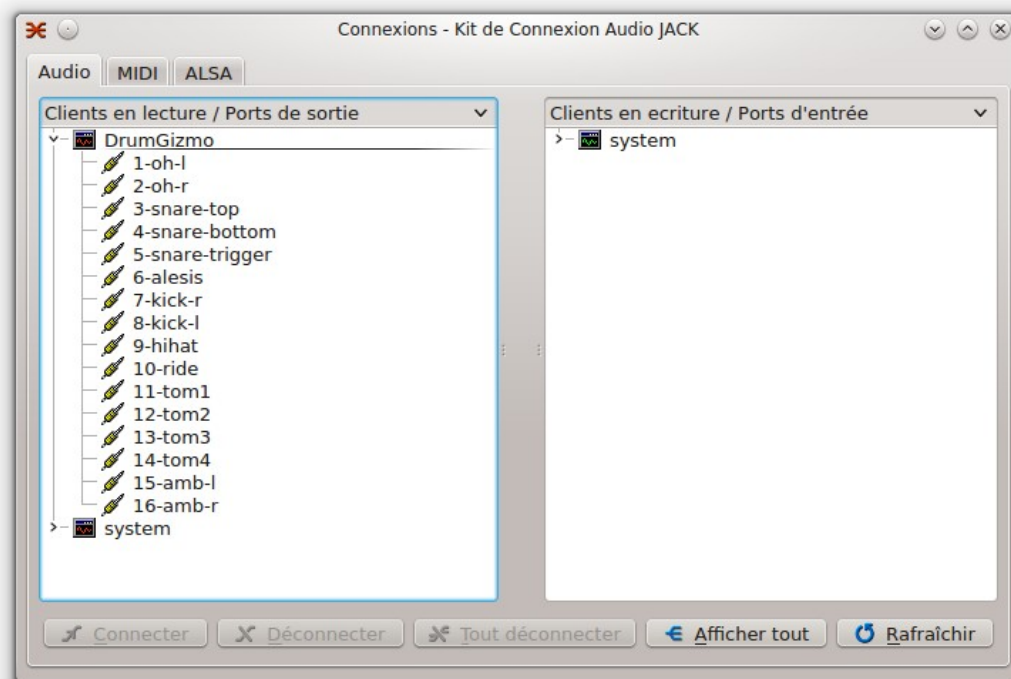


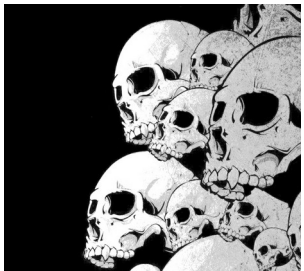
Drumgizmo

Aasimonster - Canals

Each of the microphones are mapped to an output channel when the plugin is assigned to the MIDI track. The tracks are as follows:

- Midi Ch 1: Overhead left (RØDE-NT5)
- Midi Ch 2: Overhead right (RØDE-NT5)
- Midi Ch 3: Snare top (Shure SM57)
- Midi Ch 4: Snare bottom (Shure SM57)
- Midi Ch 5: Snare trigger (DDrum)
- Midi Ch 6: Alesis DM5
- Midi Ch 7: Kick right (AKG D112)
- Midi Ch 8: Kick left (AKG D112)
- Midi Ch 9: Hihat (Cardioid condenser)
- Midi Ch 10: Rise cymbal (Cardioid condenser)
- Midi Ch 11: Tom 1 (AKG C518M)
- Midi Ch 12: Tom 2 (AKG C518M)
- Midi Ch 13: Tom 3 (AKG C518M)
- Midi Ch 14: Tom 4 / floor tom (AKG C518M)
- Midi Ch 15: Ambience left (Cardioid condenser)
- Midi Ch 16: Ambience right (Cardioid condenser)

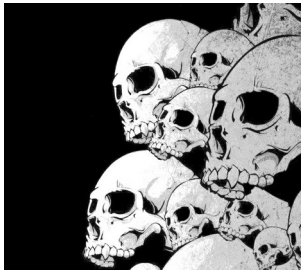




Drumkit DRSKit



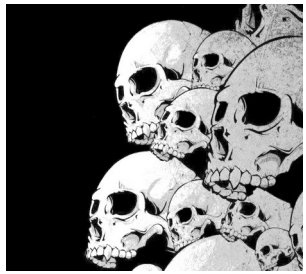
```
drumgizmo -i jackmidi -l midimap=Midimap_minimal.xml -o jackaudio DRSKit_minimal.xml
```



Drumgizmo

DRSKIT - Micros

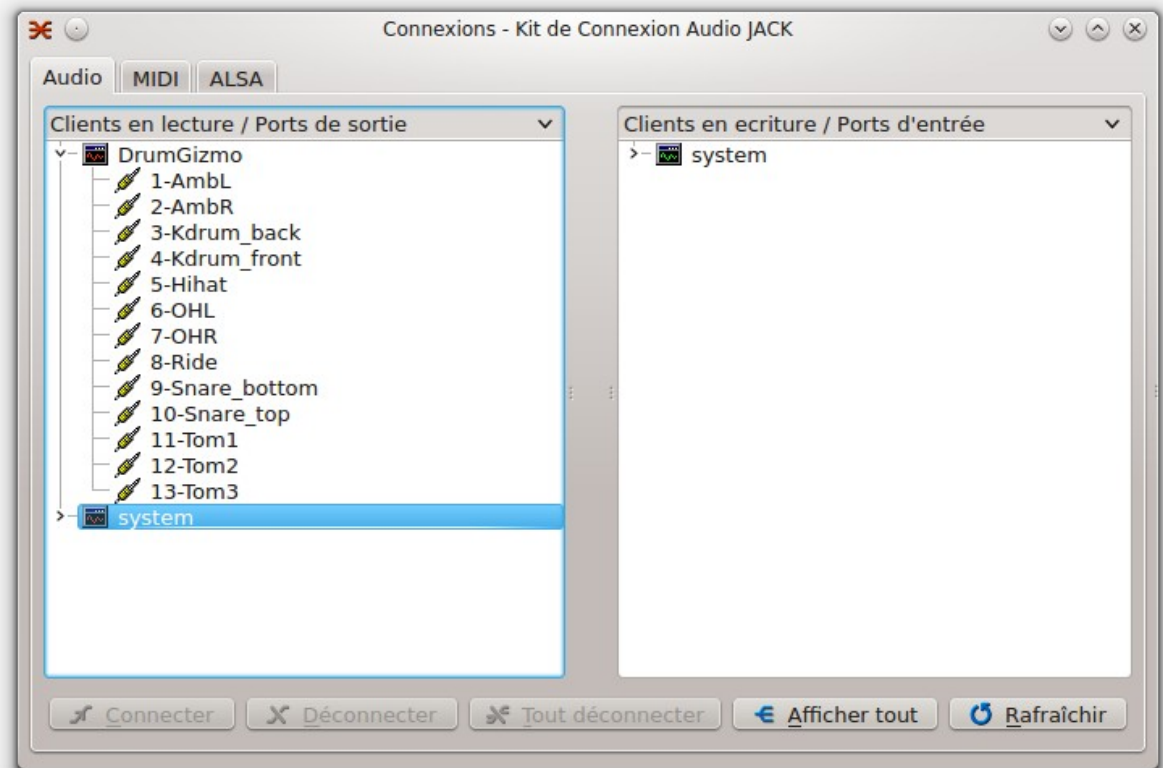
Close mic: Snare top: Shure Sm57
Close mic: Snare bottom: Shure Sm57
Close mic: Tom1: Shure Sm57
Close mic: Tom2: Shure Sm57
Close mic: Tom3: Shure Sm57
Close mic: Kick drum back: AKG D112
Close mic: Kick drum front: Shure Sm57
Close mic: Ride cymbal: Beyerdynamic MCE 86 II
Close mic: Hihat: Beyerdynamic MCE 86 II
Overhead: Focusing on left crash and hihat: Røde Nt5
Overhead: Focusing on right crash and ride: Røde Nt5
Ambience: Focusing on entire kit placed further back in the room to the left: t.bone Rb500
Ambience: Focusing on entire kit placed further back in the room to the right: t.bone Rb500

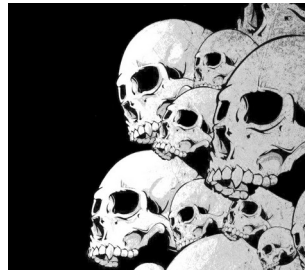


Drumgizmo DRSKIT - Canals

All microphones are connected to its own channel when loading the kit in DrumGizmo. 13 channels total. Remember to pan the relevant channels to give you a better stereo effect.

Midi Ch 1: Ambience left
Midi Ch 2: Ambience right
Midi Ch 3: Kickdrum back
Midi Ch 4: Kickdrum front
Midi Ch 5: Hihat
Midi Ch 6: Overhead left
Midi Ch 7: Overhead right
Midi Ch 8: Ride cymbal
Midi Ch 9: Snaredrum bottom
Midi Ch 10: Snaredrum top
Midi Ch 11: Tom1
Midi Ch 12: Tom2 (Floor tom)
Midi Ch 13: Tom3 (Floor tom)

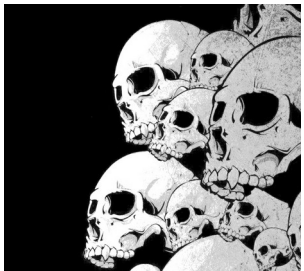




DrumKit Muldjordkit



```
drumgizmo -i jackmidi -l midimap=Midimap.xml -o jackaudio MuldjordKit.xml
```



DrumGizmo

Muldjordkit - Micros

Close mic: Snare top: Shure Sm57

Close mic: Snare bottom: Shure Sm57

Close mic: Tom1: AKG C516 ML

Close mic: Tom2: AKG C516 ML

Close mic: Tom3: AKG C516 ML

Close mic: FTom4: AKG C516 ML

Close mic: Kick drum left inside barrel: AKG D112 + DDrum trigger

Close mic: Kick drum right inside barrel: AKG D112 + DDrum trigger

Close mic: Ride cymbal left: Shure Sm57

Close mic: Ride cymbal right: Shure Sm57

Close mic: Hihat: Shure Sm57

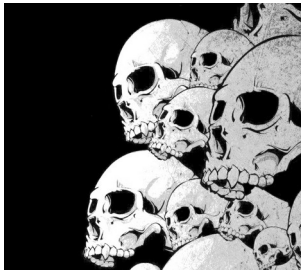
Overhead left: Røde Nt5

Overhead right: Røde Nt5

Ambience: Focusing on entire kit placed a little further back in the room to the left:
t.bone Rb500

Ambience: Focusing on entire kit placed a little further back in the room to the right:
t.bone Rb500

Also DDrum triggers on both kick drums mixed onto one mono channel

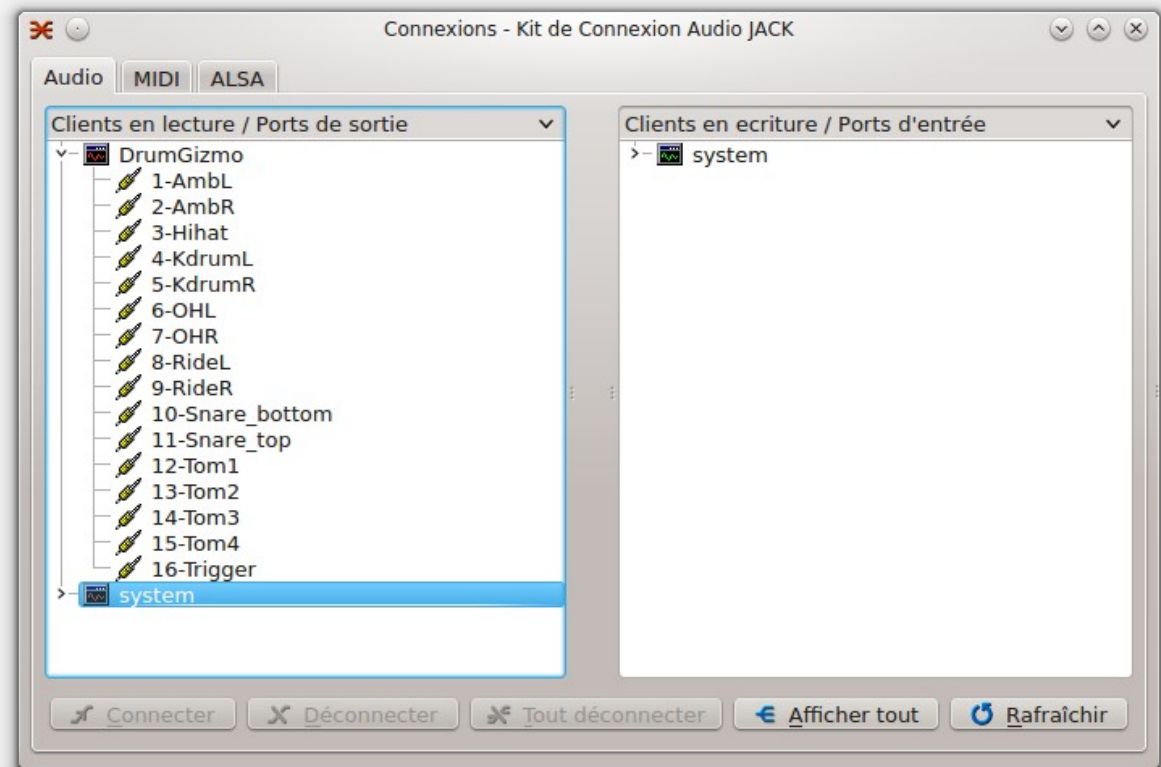


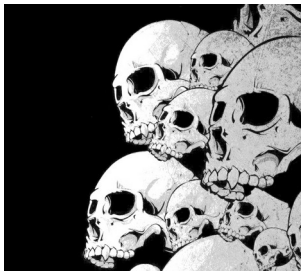
DrumGizmo

Muldjordkit - Canals

All microphones are connected to its own channel when loading the kit in DrumGizmo. 16 channels total. Remember to pan the relevant channels to give you a better stereo effect.

Midi Ch 1: Ambience left
Midi Ch 2: Ambience right
Midi Ch 3: Hihat
Midi Ch 4: Kickdrum left
Midi Ch 5: Kickdrum right
Midi Ch 6: Overhead left
Midi Ch 7: Overhead right
Midi Ch 8: Ride left
Midi Ch 9: Ride right
Midi Ch 10: Snare bottom
Midi Ch 11: Snare top
Midi Ch 12: Rack tom 1
Midi Ch 13: Rack tom 2
Midi Ch 14: Rack tom 3
Midi Ch 15: Floor tom
Midi Ch 16: Kick drum trigger signals





Hydrogen plugins



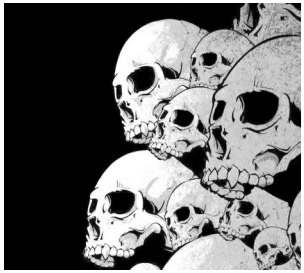
drumlabooh

drmr



drumrox





Other drum plugins

drumgizmo : Software Synthesizer

ChowKick : Kick synthesizer based on old-school drum machine circuits

boomer : A drum synth

geonkick : Drum Software Synthesizer

kickmess : A kick drum synthesizer plugin

lv2-fabla : An LV2 drum sequencer

onetrick-cryptid : An FM drum synth with the cold clanging heart of a DX7 in the fearsome frame of an 808.

onetrick-urchin : A hybrid drum synth that models the gritty lo-fi sound of beats from vintage records without sampling.

sickbeatbetty : An open source MIDI drum machine / generator VST and standalone application

AVLDrumkits-BlackPearl-H2-repack : Hydrogen drumkit

AVLDrumkits-BlondeBop-HotRod.h2d : Hydrogen drumkit

AVLDrumkits-BlondeBop.h2d : Hydrogen drumkit

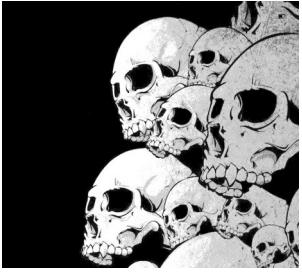
AVLDrumkits-BuskmansHoliday : Hydrogen drumkit

AVLDrumkits-RedZeppelin-H2-repack.h2d : Hydrogen drumkit

bandshed-drumkits : Bandshed DrumKits for Hydrogen

drumkv1 : An old-school drum-kit sampler

lv2-avldrums-x42-plugin : Simple Drum Sample Player LV2 Plugin



Other drum plugins

drumelidrum : drum machine VST3 plugin

drumgroovepro : A free open-source MIDI drum groove sequencer

drumlabooh : A drum machine that can use Hydrogen / SFZ and other drumkits

drumrox : A hydrogen compatible drum LV2 plugin

drummock : A simple drum sampler

drumcracker : A drum sampler compatible with drumgizmo drumkits

jdummer : An open source drum plugin that acts as an alternative to EZDrummer3